

Deep Learning as an Improved Method of Preprocessing Biomedical Raman Spectroscopy Data

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Abstract

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Keywords

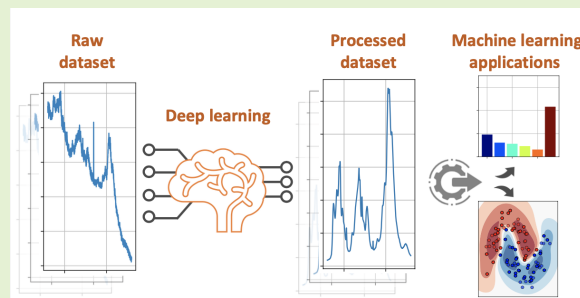
autoencoder, bioengineering, convolutional neural network, deep learning, extracellular vesicles, machine learning, raman spectroscopy, sers, signal processing and analysis

Deep Learning as an Improved Method of Preprocessing Biomedical Raman Spectroscopy Data

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Abstract—Machine learning has had a significant impact on the value of spectroscopy-based characterization tools, particularly in biomedical applications, due to its ability to detect latent patterns within complex spectral data. However, it often requires extensive data preprocessing, including baseline correction and denoising, which can lead to unintentional bias during classification. To address this, we present a deep learning-based signal preprocessing method capable of handling all the defects of raw Raman spectroscopy data without any need of human intervention. To achieve this, a novel deep convolutional neural network (CNN) architecture was trained on randomly generated spectra with various defects. We demonstrate that the proposed network results in faster training and that it can perform complete spectral preprocessing in a single step with more accuracy, speed, and defect tolerance than conventional methods. These improvements make it an ideal candidate for hyperspectral imaging applications in which tens of thousands of raw spectra may need to be processed rapidly. The superiority of this method is demonstrated for simulated Raman spectra, surface-enhanced Raman spectroscopy (SERS) imaging of chemical species, classification of low resolution Raman spectra of human bladder cancer tissue, and finally classification of SERS spectra from human placental extracellular vesicles (EVs). These findings encourage the future use of deep learning as a rapid and unbiased method of preprocessing spectroscopy data and may be particularly useful in biomedical applications involving large data sets from highly heterogeneous samples and signal defects of complex nature.

Index Terms—convolutional neural network, deep learning, extracellular vesicles, machine learning, Raman spectroscopy, SERS, Autoencoder



I. INTRODUCTION

SPECTROSCOPY is a technique that provides information regarding the atomic and molecular properties of an investigated sample based on its frequency response to optical illumination. However, the measured frequency responses, i.e. spectra, are usually contaminated by inconsistent baselines,

various types of noise, and cosmic rays. Normally, these defects must be addressed prior to accurately identifying or classifying the investigated samples. In the case of Raman spectroscopy, which is of particular interest due to its potential in noninvasive and label-free biomedical applications [1]–[4], spectral preprocessing is often unavoidable due to the inherently complex nature of the signal defects [5].

Various methods of baseline correction and denoising have been introduced to overcome spectroscopy signal defects. To some extent these methods are effective, but they suffer from the need for human intervention, which can unintentionally alter the valuable sample-specific information in the signal. For baseline correction, methods such as penalized least square [6], [7], morphological-based techniques [5], [8], polynomial fitting [9]–[11], and frequency techniques such as Fourier and wavelet transformation [12]–[14] all have some fitting parameters that need to be tuned manually. For signal denoising, methods including Savitzky–Golay (SG) [15], perfect smoother [16], smoothing based on wavelet transform [14], [17], finite impulse response [18], and smoothing based on vector casting [19] have been used previously. However, similar to baseline corrections, obtaining the optimum result

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inevitably requires human discretion and the denoised signal is always altered to some degree. Although these changes may seem negligible to the human eye, even small deviations from the original signal can have significant effects on the subsequent multivariate analysis.

Several researchers have attempted to bypass these issues entirely by using deep convolutional neural networks (CNN) to directly classify the raw Raman spectra [20]–[22]. As the training data in these methods are free from any unwanted changes caused by spectral preprocessing, they yield much higher classification accuracy and exhibit resilience against baseline changes. However, as the training of deep CNNs requires massive data sets, often exceeding tens of thousands of spectra, they are not always suitable in settings where spectral acquisition is expensive, time consuming, or simply not possible experimentally for reasons such as sample degradation. Raman spectroscopy of clinical samples suffers from this since it can be time consuming and obtaining tens of thousands or more spectra for classification purposes is rarely feasible. The other main disadvantage of raw data classification is that unlike principal component analysis (PCA) or support vector machine (SVM) classification of preprocessed spectra, in which the results are interpretable and can often be directly related to the presence or absence of some specific chemical bonds, deep CNN classification of raw data does not provide descriptive insight into the classification reasoning. This is because the extracted features in convolutional layers are unreadable, which could create major obstacles in diagnostic applications where well-defined biomarkers are preferred.

Recently, significant efforts have been made to perform signal preprocessing using variants of deep CNN [23], [24]. In one study, researchers used UNET [25], a deep CNN variant that was initially introduced for medical imaging segmentation, to automate and unify all the preprocessing stages including denoising, baseline correction, and cosmic ray removal [24]. They demonstrated the superiority of their proposed method over conventional techniques in recovering the Raman signal of several chemical species and porcine tissues. Importantly, this novel method compensated for all the drawbacks of direct classification of raw Raman signals using deep CNN, as the network training could be performed only once using simulated spectra. Then, that trained network could be used to perform rapid spectral preprocessing on new experimental data, resulting in readable and interpretable outputs. However, their proposed UNET method cannot perform denoising and baseline correction independently. Therefore, standalone denoising or baseline correction, for the direct comparison of their method to conventional techniques, was not possible. This also limits its application in the rapidly growing field of SERS characterization of extracellular vesicles (EVs), [26]–[28] in which the produced signals vary greatly in intensity, and recovering low signal to noise ratio (SNR) spectra can generate false peaks. This issue can potentially be avoided by having access to the signal that has only undergone baseline correction prior to denoising, but would require the development and validation of a new deep CNN preprocessing architecture.

In this study, we propose the use of a deep CNN and

autoencoder-based network architecture for the fully automated preprocessing of Raman spectra in a diverse range of applications. In contrast to the previously developed deep CNN approaches, the proposed network is uniquely capable of offering full access to both the baseline corrected and denoised spectra, allowing its direct comparison to conventional preprocessing methods. The proposed network is trained using simulated spectra and then applied to experimental test spectra from several sources, including low resolution Raman spectra from human bladder cancer tissue, and hyperspectral SERS imaging of chemical species and placental EVs from tissue explant cultures. The success of this approach will encourage further exploration of using deep CNNs to preprocess spectroscopy data in EV SERS and other biomedical applications as it can remove human bias while still providing full access to the data, ultimately providing a means to improve rigor and reproducibility.

II. METHODS

A. Baseline Correction

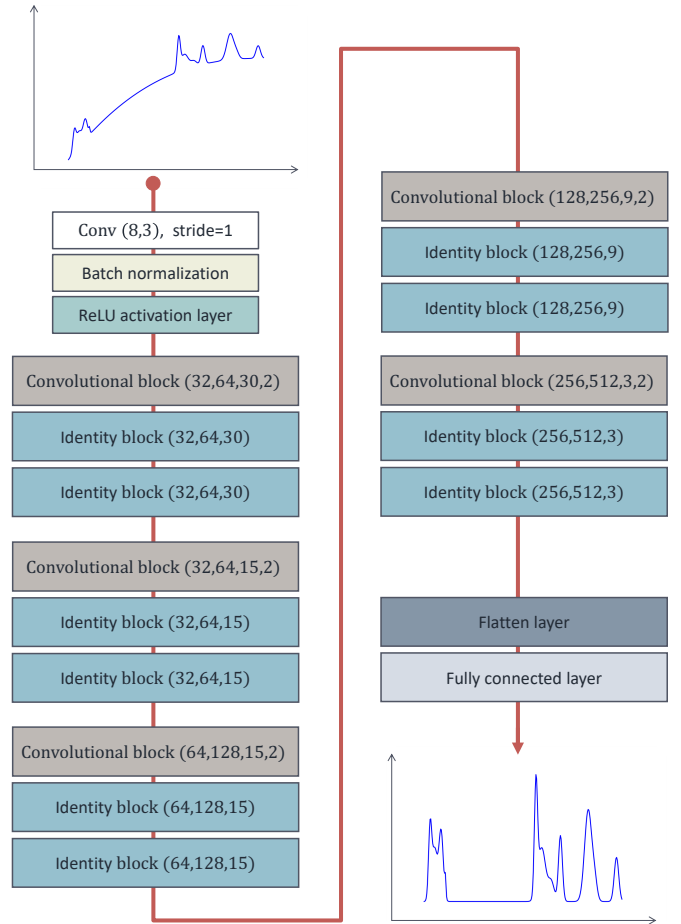


Fig. 1. Proposed architecture for spectral baseline correction. Each of the convolutional and identity block consists of stack of three convolution layers to achieve a sufficiently deep network, as shown in Figure 2.

First, we designed a network specifically for spectral baseline correction. In order to effectively handle complex baselines, the designed network must have sufficient depth. However, stacking multiple convolution layers results in two main

issues for the network training, namely vanishing gradient and accuracy saturation and degradation. Residual neural network (ResNet) [29], as an architecture of deep CNN, has solved these two problems by introducing skips which bypass some of the in between layers, allowing the backpropagated error to reach deeper layers toward the network's input. This results in much higher accuracy in several applications, including the classification of raw Raman spectra [20], [22]. Therefore, the network illustrated in the Figure 1, with ResNet-inspired skips implemented in the convolutional and identity blocks, is proposed for baseline correction. The schematics of these blocks are shown in the Figure 2, where the notations n_1^i and n_2^i are the number of filters in the different one dimensional convolution layers (CONV), and F^i and s^i are the size and stride step defined for the filters, respectively. The reason that two different blocks are defined as having skips is to be able to change the number of filters, their sizes, and strides through the network. For example, strides lead to shrinkage of the feature map size of the output and can cause mismatches between the shape of skip outputs and the main routes. To avoid this, we added a convolutional layer with strides in the skip routes of the convolutional blocks. We have also used batch normalization layers to first regularize the model and mitigate overfitting, as well as to allow us to be less dependent on the network initialization [30].

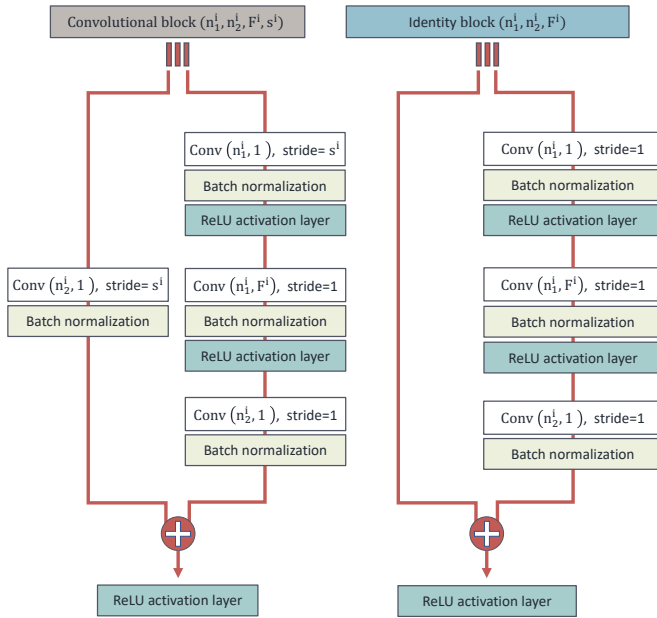


Fig. 2. Schematic of convolutional and identity blocks used in Figure 1, showing the skip connections which bypass some convolution layers. Due to a stride used in the convolutional block, there is a convolution layer in its skip route to match its output shape to the output of its main route.

The network was trained with 120,000 randomly generated artificial Raman spectra. Each spectrum consisted of Gaussian peaks with random amplitudes (between 100 to 1000), variances (.0025 to .025 of the wave-number span), and centers (uniformly chosen across the wave-number span). Baselines were added to the generated spectra using randomly generated

polynomial functions to a maximum degree of 7. For the network training, batch sizes of 64, a learning rate of 2×10^{-4} , and a kernel regularizer of 12 (with regularization parameters equal to .001) were chosen for all one dimensional convolution layers to avoid overfitting. Multiple loss functions, including Mean Absolute Error (MAE), Mean Squared Error (MSE), the logarithm of hyperbolic cosine (LogCosh), and Cosine Similarity (CS) were selected to evaluate their effects over the training and final accuracy.

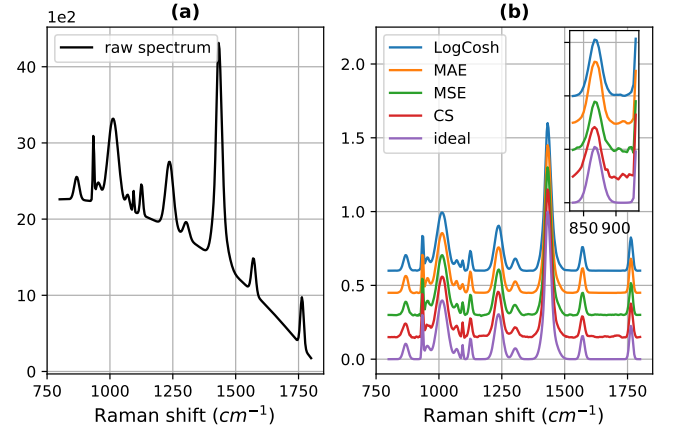


Fig. 3. Spectral baseline test showing (a) an example Raman spectrum in the test set and (b) the output of the proposed deep learning method with LogCosh, MAE, MSE, and CS losses compared to its ideal output.

Accuracy and validation loss for each of the networks over 100 training epochs are shown in Supplementary Figure 1. Clearly, the network with the CS loss function achieved very low loss much faster than networks with other loss functions. This occurred due to the sensitivity of CS loss to the positions of the peaks which is the inherent nature of spectroscopy signals. However, as in the calculation of the CS loss function, the amplitude of the output signal is not considered, and information regarding the total energy of the signal is lost. The output of each of these networks for one spectrum in the test set when trained with various loss functions is shown in the Figure 3. For the test set, Gaussian peak parameters of a wider range than those in the training set were used to test the generalization of the trained network.

To better compare our deep learning approach with conventional techniques, we generated a test set similar to the training set with varying baseline complexities. To add baselines to the test set, polynomial degrees between 0 to 20 were chosen, and for each degree of polynomial, 1000 random spectra were generated leading to a total of 21000 spectra in the test set. Removal of the added baselines was then attempted using improved morphological method (iMor), asymmetric least square smoothing (AsLS), adaptive iterative penalized least square smoothing (airPLS) methods, and our deep learning approach. The result is depicted in the Figure 4 with various hyperparameter considerations for iMOR (half window sizes between 1 to 20), AsLS (smoothing factor between 250 to 2.5×10^8), and airPLS (smoothing factor between 250 to 2.5×10^8). Clearly, our the deep learning method performed

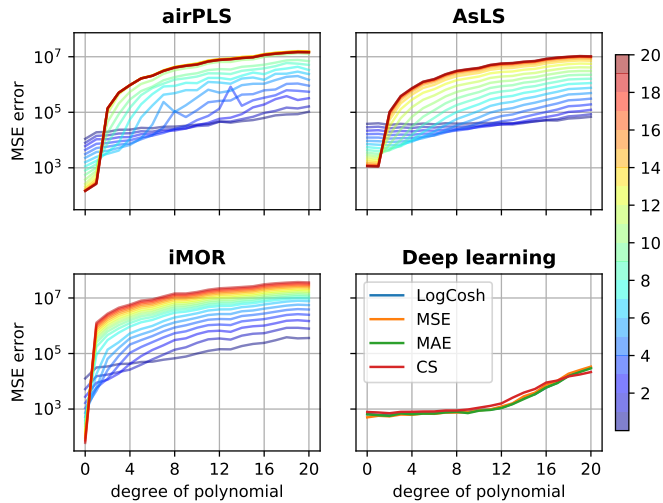


Fig. 4. MSE of conventional baseline correction techniques including adaptive iterative penalized least square smoothing (airPLS), asymmetric least square smoothing (AsLS), and improved morphological method (iMOR) methods with various tuning hyperparameters compared to the proposed deep learning methods with various losses for baselines of various polynomial degrees.

much better for most baseline complexities compared to conventional methods, regardless of the loss function or tuning hyperparameters used.

B. Denoising

Similar to the baseline correction, we used the network depicted in Supplementary Figure 2, which includes skips that bypass some of the convolutional layers. Training spectra were again generated similar to the baseline correction section, using random Gaussian functions with various centers, variances, and amplitudes while Gaussian noise (with a mean of zero and variance randomly chosen between zero and 100) was added proportional to the minimum amplitude of the generated peaks. CS, MSE, MAE, and LogCosh loss functions were used to train the network, and once again, CS yielded the fastest training but did not preserve any information regarding the total energy of each signal. Training and validation loss for 100 epochs is also shown in Supplementary Figure 3. Two spectra from the test set with different signal to noise ratio (SNR) levels and their corresponding denoised spectra after deep learning preprocessing are also presented in Figure 5.

To show the superiority of the deep CNN over conventional methods, 21000 spectra were again generated for the test set. With each consecutive batch of 1000 spectra, the amount of Gaussian noise was increased continually, with variance from zero up to two times the minimum peak height within each respective spectrum. The noise was removed using SG (window length of 21 and polynomial order of 2, 4, 6, 8, and 10), wavelet denoising (BayesShrink algorithm with symlets 8 wavelet and wavelet levels between 2 to 20), and then compared with denoising methods obtained using the proposed deep learning method. As shown in Figure 6, deep learning again performed much better than any conventional denoising methods, regardless of how their hyperparameters

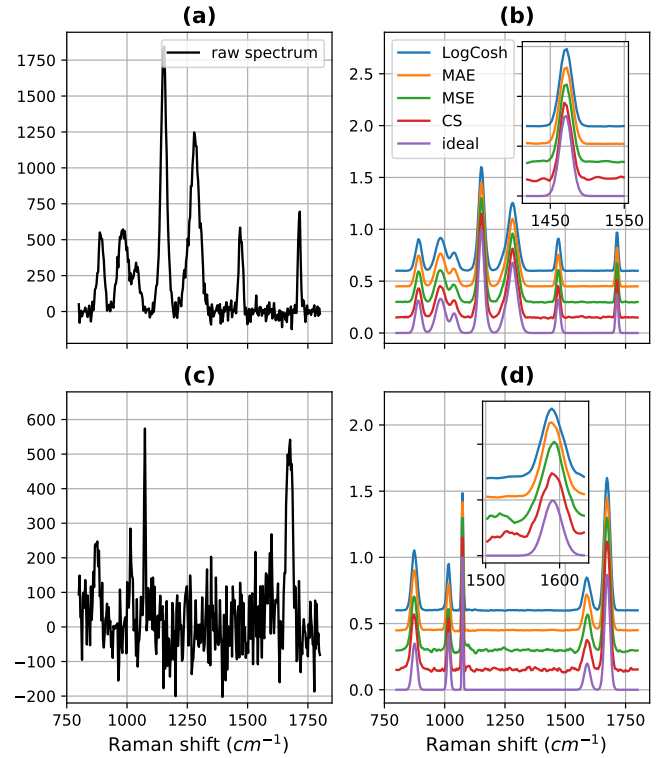


Fig. 5. Spectral denoising test showing (a) and (c) example spectra from the denoising test set with high and low SNR, respectively, (b) and (d) the output of the proposed deep learning methods trained with various losses compared to the ideal denoised spectra when the deep learning methods are fed by (a) and (c), respectively.

were tuned. Please note that the number following the SG in the figure legend denotes the polynomial order chosen for the SG algorithm. Additionally, only the best WL result is presented, which was achieved using a wavelet level of 18. Clearly, increasing the polynomial order in SG method improves the result for high SNR spectra (lower variance of the Gaussian noise, i.e. left side of the horizontal axis in Figure 6) while it results in weaker performance for the low SNR spectra (higher variance of the Gaussian noise, i.e. right side of the horizontal axis in Figure 6).

C. Baseline Correction, Denoising, and Cosmic Ray Removal

To perform baseline correction, denoising, and cosmic ray removal at the same time using deep learning, we first generated an appropriate training data set. This was done by adding Gaussian noise and random spikes (synthetic cosmic rays) to the data originally generated for the baseline correction development. However, to cover a greater variety of the Raman peak shapes and increase the versatility of the network, we also used Voigt, Lorentzian, and Moffet functions in addition to the previous Gaussian function to form peaks in the generated spectra. This ensures that the network is trained in a way which can handle more peak shapes and outputs, enabling the preservation of the actual signal. Subsequently, we increased the training set to 240000 spectra while the batch size of

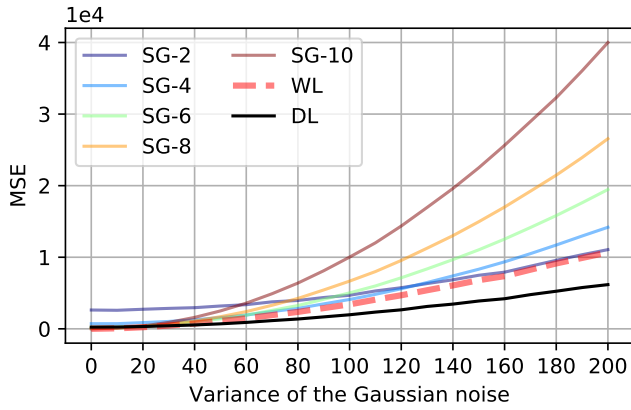


Fig. 6. Performance comparison (MSE) of conventional spectral denoising techniques with the proposed deep learning method for increasing Gaussian noise variance.

training and learning rate is increased to 256 and 4×10^{-4} , respectively. As this is a relatively complex task, we designed multiple cascading network architectures to be able to compare and select the best possible approach. The first proposed architecture progresses from the baseline correction network to the denoising network presented in the previous sections, as shown in the Figure 7(a).

The other method is obtained by utilizing the notion of an autoencoder. Autoencoders are a subclass of neural network that aim to replicate their input at their output after passing through a hidden layer. Such networks consist of three major parts: an encoder, a latent space, and a decoder. Usually, the encoder section compresses the dimensionality of the data and feeds it to the lower dimensional layer, i.e. the latent space. Then, the dimensionality of the latent space is expanded to the original input dimensions using the decoder. These networks have been used extensively in applications including non-linear embedding or non-linear dimensionality reduction and the development of denoising techniques. Ideally, the reduction of the input's dimensionality does not leave enough room for noise such that the trained network's output will be almost noise free. In proposed network, as the output is not identical to its input, it is not considered a conventional autoencoder. However, the same idea of dimensional reduction in the middle of the network holds. Assuming that a bottleneck in the latent space will not leave enough room for the baselines and noise, which have a random nature, they should be rejected as the network becomes more trained. The schematic of this proposed network is presented in the Figure 7(b).

The other main advantage of having a latent layer in the middle of the network is to avoid the extreme reduction of the output's feature map at the last layer of the decoder (in the case of directly cascading the baseline correction and denoising networks). Although strides reduce the computational complexity and can significantly increase the training speed, they also shrink the feature map of the output's tensor by a factor of two (due to a stride of 2 being used in the convolutional block). Therefore, having too many convolutional blocks with strides, to achieve a deeper network, exponentially reduces the

final feature map and may lead to weak reconstruction of the spectra at the network's output. In addition, in the extreme case of a very deep network, this can also lead to an output width of 1 in the middle of the network, making the rest of the convolutional layers practically ineffective. Thus, including the latent layer in our network expands the feature map and allows a deeper network.

For the final method, we propose choosing the dimension of the latent space equal to that of the input and considering the latent space as another output of the network, as shown in the Figure 7(c). In this case, it can also participate in the supervised network training by considering its appropriate output goals. For example, we can assume the noisy spectra without the baseline as the ideal output for the latent space. Therefore, the encoder will be trained to remove the baseline while the final output at the end of the decoder will train both encoder and decoder for noise reduction. This is particularly true, as the generated errors at each output layer can only propagate backwards through the network. In this case, the denoising and baseline correction sections will be adaptive while performing their tasks.

The training and validation loss over 120 epochs of training for the all proposed networks is depicted in the Supplementary Figure 4. Clearly, the network with an extra output at its latent layers outperformed all the proposed networks in terms of both training speed and the final achieved training and validation losses. Furthermore, increasing the size of the latent layer in the second scenario leads to improvement of both training speed and achieved training and validation loss.

III. RESULTS AND DISCUSSION

To show the performance and superiority of our proposed method we use three different examples, all of which would have previously required extensive preprocessing. Our first example is the Raman hyperspectral imaging of a chemical droplet with a very low concentration released over the boundary of a previously patterned SERS substrate (known as LINST) [28]. This is considered a difficult task for conventional preprocessing techniques as it deals with signals of various amplitudes and baselines, and contains tens of thousands of spectra such that manual tuning of hyper parameters specifically for each spectrum is practically impossible.

The second example includes low resolution Raman spectra of bladder cancer tissue [4]. Low cost, low resolution, and portable Raman spectrometers make the use of Raman spectroscopy feasible in the clinical environments. However, these systems suffer from wide range of problems. First, the spectra they obtain suffer from severe noise levels as they are usually not equipped with cooled CCDs and therefore thermal noise can easily overwhelm the weak Raman spectra of the sample. Their second major drawback is the inconsistency of baselines in the acquired spectra. We previously demonstrated that any change in the baseline of Raman spectra can lead to false classification of the sample [22]. For this reasons, an ideal preprocessing method needs to be able to remove many different baselines without creating artifacts that bias the subsequent processing steps.

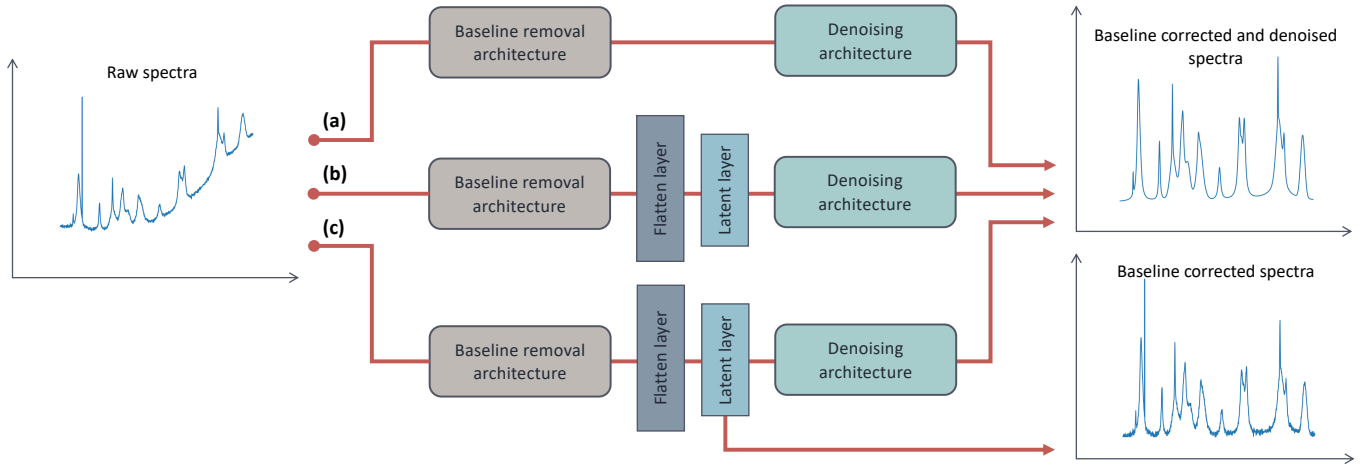


Fig. 7. Cascaded deep CNN architectures for complete spectral preprocessing. (a) First network architecture scenario: Baseline fitting followed by noise removal. (b) Second network architecture scenario: Baseline fitting is terminated to a latent layer to compress the dimensions and expand the features map size. The added latent layer then followed by the denoising architecture to produce the output. (c) Third network architecture scenario: The latent layer at the middle of network is also producing an output and trained via noisy signal.

The third and most complex example includes SERS spectra of placental EVs from tissue explant cultures [28]. As previously mentioned, EV SERS spectra are highly diverse due to the heterogeneous cargo of each EV and their placement relative to the SERS surface. As a result, manual spectral preprocessing is often required in order to handle the wide range of noises and baselines.

A. Hyperspectral Raman Imaging of Chemical Species

For this example, 10,000 spectra from a dried Rhodamine 6G droplet (20 μL) on top of a rectangular patterned SERS substrate were collected using a 532nm laser with 100mW power and only 100ms (.1s) signal acquisition at each point. These spectra were obtained over a grid of 100×100 points and were used to create a 2D image of the surface. The SERS substrate used in this example was fabricated by sputter deposition of gold onto a silicon wafer, followed by a femtosecond laser-machining process of a rectangular grid (1 mm x 1.5 mm) to create randomly distributed SERS hotspots [28]. Therefore, this data set contains various signal intensities and SNRs, requiring extensive care when preprocessing using conventional methods.

2D images of the surface were created by considering the peak height of the Raman signal at $1648(\text{cm}^{-1})$ after each processing method. These images are shown in the Figure 8 for the airPLS, second deep learning scenario (with a latent size of 500), and third deep learning scenario. Both deep learning methods outperformed the conventional approach as they lead to sharper images with much more detail. The peaks' heights are also more preserved in the both deep learning methods as the maximum peak in second and third deep learning methods are 1077 and 1237 cm^{-1} , respectively, while it is 768 cm^{-1} in the data processed by airPLS method. This change in the location of the maximum peak height location can be caused by inappropriate tuning of baseline correction algorithms, and is one example of how a user may unintentionally bias the data while performing preprocessing.

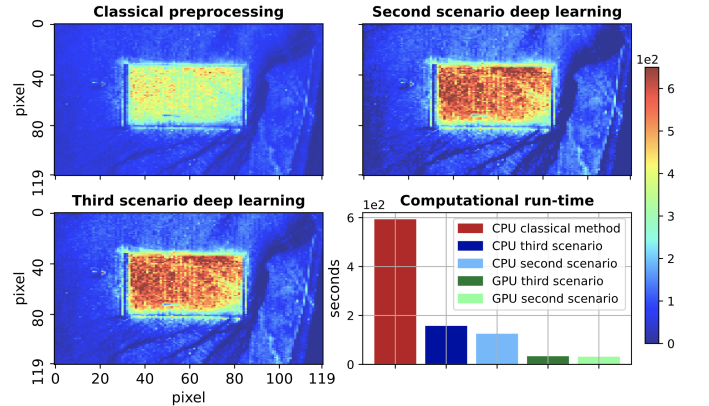


Fig. 8. The imaging of the height distribution of 1648cm^{-1} peak of R6G Raman spectra dried on the SERS surface when (a) all spectra are preprocessed using conventional methods, (b) second deep learning scenario, and (c) the third deep learning scenario. (d) The computation run-times are also shown for the conventional method and proposed deep learning methods for both CPU and GPU.

The other main advantage of the proposed deep learning methods are their computational speed. Despite the long computational times required for training, they are much faster than the conventional method when applied to new experimental data. Moreover, their algorithms are inherently GPU friendly and their computation can take the advantage of hyperthreading by GPUs. The computational speed of the conventional methods compared to the proposed deep learning approaches using both CPU and GPU are shown in Figure 8d. While the proposed methods are more than 4 times faster using the CPU (Intel Core i3 1000NG4) compared with the conventional methods, they are 20 times faster using the GPU (NVIDIA Quadro P5000).

B. Low Resolution Raman Spectroscopy of Bladder Cancer Tissue

In this example we used previously obtained Raman spectra of bladder cancer tissues using a low resolution (36 cm^{-1})

portable Raman spectroscopy system [4]. This data contains 2592 spectra obtained using various laser powers (30-150 mW) and each was classified as either coming from either healthy tissue or low or high-grade tumours. As previously described, such data suffers from high levels of noise and inconsistency of the baselines, requiring extensive care during preprocessing. This data was divided into two subsets of training and test sets for the purpose machine learning classification. The PCA transformation of the training dataset after it is processed by AsLS combined with WL denoising, as well as processing using the proposed deep learning algorithm are shown in Figure 9. Clearly, the data which was processed using the our deep learning method (Fig. 9a) is better separated than that done using the conventional method (Fig. 9b).

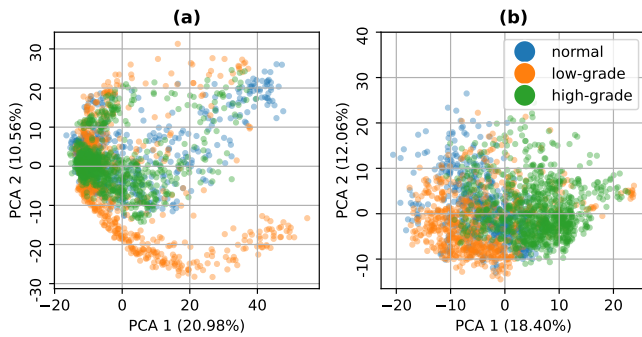


Fig. 9. Two dimensional PCA projection of the low resolution bladder cancer data set when the preprocessing was carried out by (a) the proposed deep learning and (b) AsLS combined with WL denoising. The explained variance of each principal component is shown next to its corresponding axis.

Next, we compared the performance of conventionally preprocessed data combined with conventional machine learning classifiers with our previously established deep CNN raw data classifier as shown in in Figure 10a with varying degrees of baseline alteration (detailed methods provided in supplementary information). Clearly, conventional preprocessing combined with conventional machine learning underperforms compared to deep CNN classification using only raw data as baseline complexity increases. Additionally, we compared the same deep CNN raw data classification with the proposed deep learning preprocessing combined with conventional machine learning classifiers as shown in Figure 10b. Clearly, our proposed deep learning preprocessing method not only improves the performance of conventional machine learning classifiers, but elevates their performance in most cases to the level of deep CNN raw data classification, even when data is augmented with baselines of high complexity. However, unlike deep CNN raw data classification, the preprocessing step allows to explicit examination of the spectra and the identification particular Raman bands of interest corresponding to biochemical differences between the samples. Thus, our two step method is preferable for clinical applications where understanding the molecular basis for sample classification or biomarker discovery is a focus of the study.

Other advantages of the proposed method are the computational speed of classification and its usability with smaller

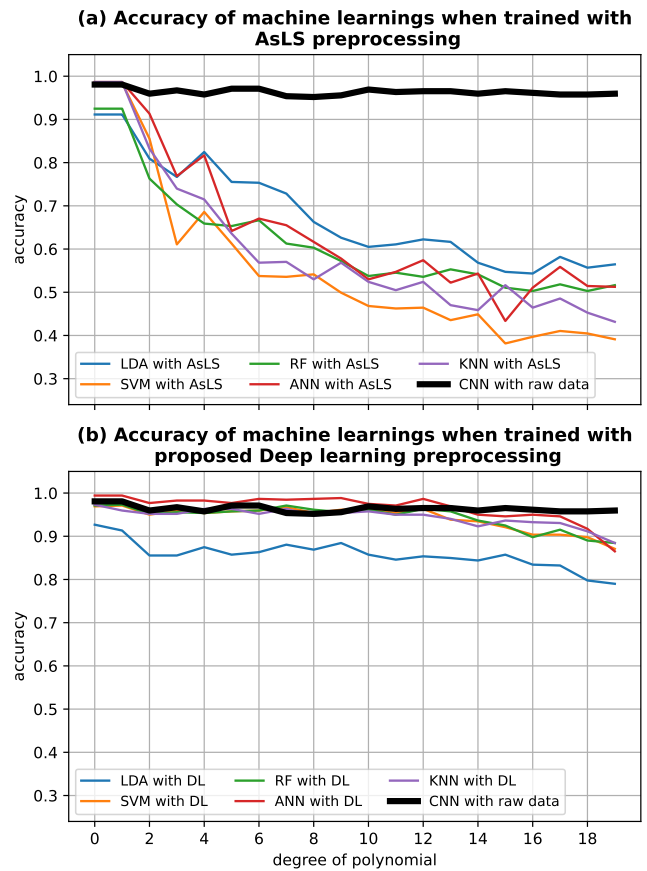


Fig. 10. Accuracy of trained machine learning methods for the altered test set using various polynomial baselines. (a) Altered test set is pre-processed by AsLS and WL. (b) Altered test set is preprocessed using proposed deep learning method. For conventional machine learning, we used methods including linear discriminant analysis (LDA), support vector machine (SVM) with RBF kernel, K nearest neighbour (KNN) and, multilayer perceptron as a classical artificial neural network (ANN).

datasets. While the raw data CNN classification architecture performs well [22], it requires relatively large datasets, and one may even need to augment datasets by up to 10 fold. However, even augmentation cannot circumvent this issue when datasets are very small, as is the case when samples are either prohibitively expensive or rare. This is particularly true because augmenting the data more than 10 fold results in heavily biased data and does not necessarily help the generalization of the network for use with the test set. Finally, training this type of raw data CNN classification network over naturally large or heavily augmented data sets requires hours of training, in clear contrast to the few seconds required to apply the trained deep learning preprocessing method and standard machine learning classification approaches.

C. SERS Spectra of Placental EVs

The final example involves the SERS spectra of normotensive and preeclamptic placental EVs from tissue explant cultures. As before, various signal amplitudes and SNR are acquired due to two primary factors. First, smaller EVs likely produce stronger Raman signals than larger ones due to their greater statistical likelihood of settling near nanoscale SERS

hotspots. Second, the biochemical contents of EVs are very heterogeneous, and even SERS spectra of EVs produced from a single cell line can be quite different. These two factors make the classifying EV from their SERS spectral data a highly challenging problem. Our fully automated method can effectively mitigate these issues as seen in Figure 11. Here, two raw SERS spectra obtained from placental EVs using identical acquisition parameters are shown in Figure 11a-b, and the preprocessed spectra are shown in Figure 11c. Clearly, our preprocessing technique successfully extracted the weaker signal and it can be seen that its peaks match those of the stronger spectrum, despite their significant differences in SNR and baselines.

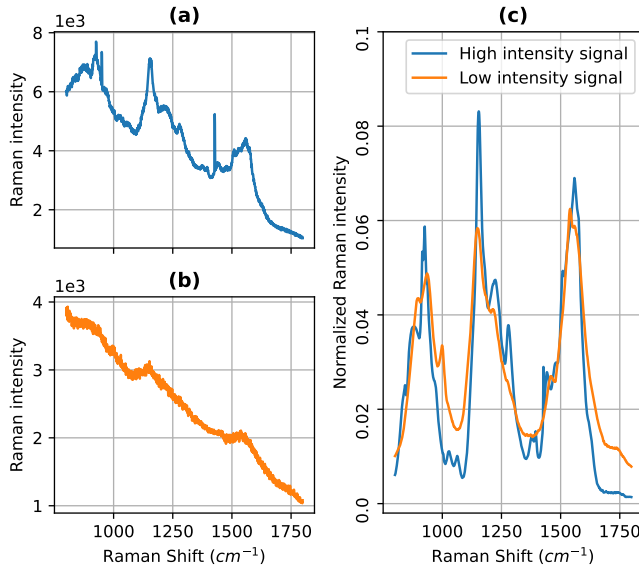


Fig. 11. Two examples of the spectra in the placental EV SERS dataset showing (a) a spectrum with high SNR, (b) a spectrum with low SNR, and (c) the preprocessed results of the spectra depicted in (a) and (b) using the proposed deep learning method.

As with the bladder cancer tissue example, various machine learning and deep learning techniques were tested for classifying the EVs based on their SERS spectra (both preprocessed and raw spectra). The results are shown in Figure 12, demonstrating that for all methods, preprocessing of the data using the proposed deep learning method improves the final classification performance compared AsLS preprocessing. The resulting accuracy is also comparable to the results from a state of art deep CNN using raw data and a bottleneck classifier (BC) [22], [28]. Similar to the bladder cancer tissue data set, the proposed method achieved this while also reducing the training time by several orders of magnitude and provided interpretable spectral information for more specific biochemical comparisons. This is because the conventional classification methods are not trained using an augmented data, which is required for the other methods based on deep CNN.

IV. CONCLUSION

In this study, we developed an improved deep learning solution for the signal preprocessing of Raman spectroscopy

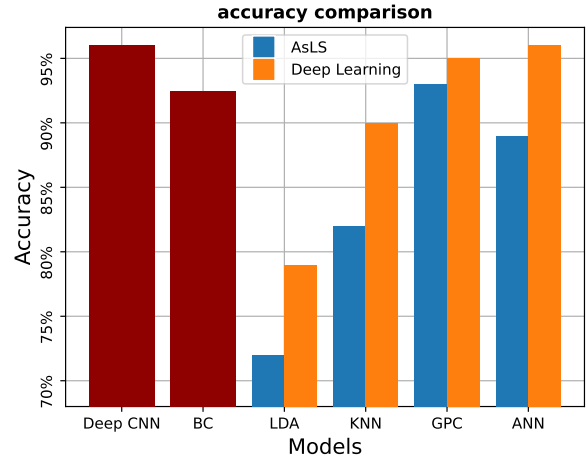


Fig. 12. Comparison of deep CNN classification of raw data and conventional machine learning classification accuracy. Data for conventional machine learning classification are shown when preprocessing is done by either AsLS or the proposed deep learning preprocessing.

data, including baseline correction, denoising, and cosmic ray removal. The superiority of the proposed method in terms of both performance and computational time are demonstrated through various examples including synthetic spectra, large data sets from Raman imaging of chemical species, and biological samples including bladder cancer tissue and placental EVs. We also showed that all conventional preprocessing methods can statistically bias the data when severe defect scenarios exist, and that this can significantly effect the classification accuracy of conventional machine learning methods. The proposed method also increases the accuracy of conventional machine learning algorithms to the level of state of art deep learning classification on the raw spectra, while also providing access to the preprocessed data for specific biochemical comparisons. Ultimately, this robust approach should enhance the value of Raman spectroscopy, particularly in biomedical applications, where large data sets containing diverse signal defects must be rapidly preprocessed, and where removing the potential of unintentional user bias will help to improve rigor and reproducibility.

V. ACKNOWLEDGEMENTS

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REFERENCES

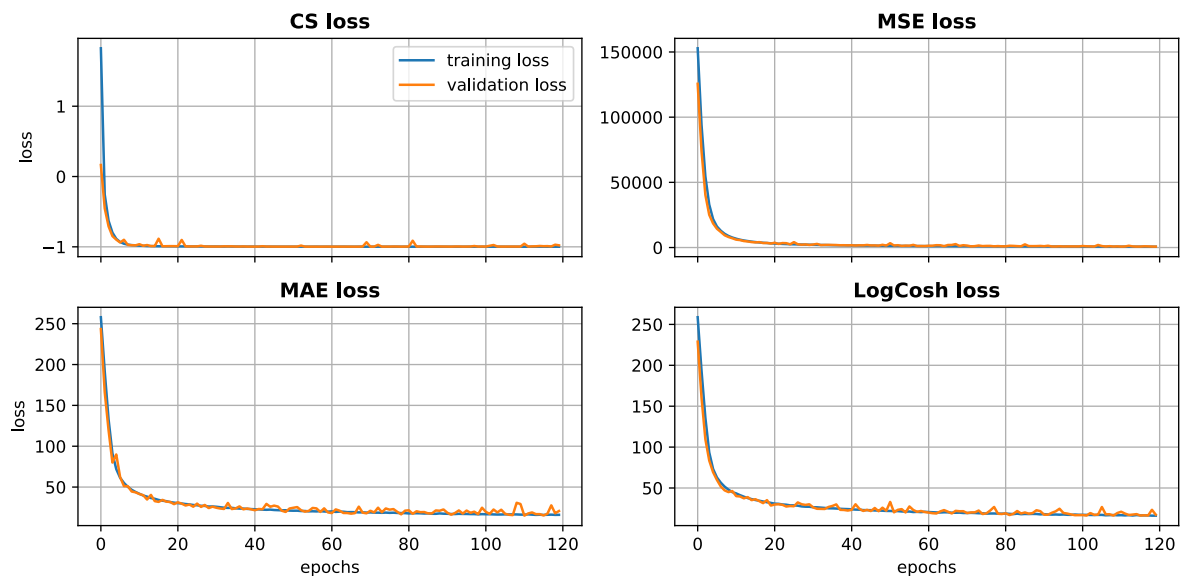
- [1] A. S. Haka, K. E. Shafer-Peltier, M. Fitzmaurice, J. Crowe, R. R. Dasari, and M. S. Feld, "Diagnosing breast cancer by using raman spectroscopy," *Proceedings of the National Academy of Sciences*, vol. 102, no. 35, pp. 12 371–12 376, 2005.

- [2] M. Vendrell, K. K. Maiti, K. Dhaliwal, and Y.-T. Chang, "Surface-enhanced raman scattering in cancer detection and imaging," *Trends in biotechnology*, vol. 31, no. 4, pp. 249–257, 2013.
- [3] Z. Movasaghi, S. Rehman, and I. U. Rehman, "Raman spectroscopy of biological tissues," *Applied Spectroscopy Reviews*, vol. 42, no. 5, pp. 493–541, 2007.
- [4] H. Chen, X. Li, N. Broderick, Y. Liu, Y. Zhou, J. Han, and W. Xu, "Identification and characterization of bladder cancer by low-resolution fiber-optic Raman spectroscopy," *Journal of Biophotonics*, vol. 71, p. e201800016, Jun. 2018.
- [5] Y. Chen and L. Dai, "An automated baseline correction method based on iterative morphological operations," *Applied spectroscopy*, vol. 72, no. 5, pp. 731–739, 2018.
- [6] J. Peng, S. Peng, A. Jiang, J. Wei, C. Li, and J. Tan, "Asymmetric least squares for multiple spectra baseline correction," *Analytica chimica acta*, vol. 683, no. 1, pp. 63–68, 2010.
- [7] Z.-M. Zhang, S. Chen, and Y.-Z. Liang, "Baseline correction using adaptive iteratively reweighted penalized least squares," *Analyst*, vol. 135, no. 5, pp. 1138–1146, 2010.
- [8] R. Perez-Pueyo, M. J. Soneira, and S. Ruiz-Moreno, "Morphology-based automated baseline removal for raman spectra of artistic pigments," *Applied spectroscopy*, vol. 64, no. 6, pp. 595–600, 2010.
- [9] P. Cadusch, M. Hlaing, S. Wade, S. McArthur, and P. Stoddart, "Improved methods for fluorescence background subtraction from raman spectra," *Journal of Raman Spectroscopy*, vol. 44, no. 11, pp. 1587–1595, 2013.
- [10] J. Liu, J. Sun, X. Huang, G. Li, and B. Liu, "Goldindc: a novel algorithm for raman spectrum baseline correction," *Applied spectroscopy*, vol. 69, no. 7, pp. 834–842, 2015.
- [11] C. Gallo, V. Capozzi, M. Lasalvia, and G. Perna, "An algorithm for estimation of background signal of raman spectra from biological cell samples using polynomial functions of different degrees," *Vibrational Spectroscopy*, vol. 83, pp. 132–137, 2016.
- [12] Y. Hu, T. Jiang, A. Shen, W. Li, X. Wang, and J. Hu, "A background elimination method based on wavelet transform for raman spectra," *Chemometrics and Intelligent Laboratory Systems*, vol. 85, no. 1, pp. 94–101, 2007.
- [13] Y. Liu, W. Cai, and X. Shao, "Intelligent background correction using an adaptive lifting wavelet," *Chemometrics and Intelligent Laboratory Systems*, vol. 125, pp. 11–17, 2013.
- [14] P. M. Ramos and I. Ruisánchez, "Noise and background removal in raman spectra of ancient pigments using wavelet transform," *Journal of Raman Spectroscopy: An International Journal for Original Work in all Aspects of Raman Spectroscopy, Including Higher Order Processes, and also Brillouin and Rayleigh Scattering*, vol. 36, no. 9, pp. 848–856, 2005.
- [15] A. Savitzky and M. J. Golay, "Smoothing and differentiation of data by simplified least squares procedures," *Analytical chemistry*, vol. 36, no. 8, pp. 1627–1639, 1964.
- [16] P. H. Eilers, "A perfect smoother," *Analytical chemistry*, vol. 75, no. 14, pp. 3631–3636, 2003.
- [17] F. Ehrentreich and L. Sümmchen, "Spike removal and denoising of raman spectra by wavelet transform methods," *Analytical chemistry*, vol. 73, no. 17, pp. 4364–4373, 2001.
- [18] M. Člupek, P. Matějka, and K. Volka, "Noise reduction in raman spectra: Finite impulse response filtration versus savitzky–golay smoothing," *Journal of Raman Spectroscopy: An International Journal for Original Work in all Aspects of Raman Spectroscopy, Including Higher Order Processes, and also Brillouin and Rayleigh Scattering*, vol. 38, no. 9, pp. 1174–1179, 2007.
- [19] M. T. Gebrekidan, C. Knipfer, and A. S. Braeuer, "Vector casting for noise reduction," *Journal of Raman Spectroscopy*, vol. 51, no. 4, pp. 731–743, 2020.
- [20] J. Liu, M. Osadchy, L. Ashton, M. Foster, C. J. Solomon, and S. J. Gibson, "Deep convolutional neural networks for raman spectrum recognition: a unified solution," *Analyst*, vol. 142, no. 21, pp. 4067–4074, 2017.
- [21] W. B. Sohn, S. Y. Lee, and S. Kim, "Single-layer multiple-kernel-based convolutional neural network for biological raman spectral analysis," *Journal of Raman Spectroscopy*, vol. 51, no. 3, pp. 414–421, 2020.
- [22] M. Kazemzadeh, C. L. Hisey, K. Zargar-Shoshtari, W. Xu, and N. G. Broderick, "Deep convolutional neural networks as a unified solution for raman spectroscopy-based classification in biomedical applications," *Optics Communications*, p. 127977, 2022.
- [23] J. Wahl, M. Sjö Dahl, and K. Ramser, "Single-step preprocessing of raman spectra using convolutional neural networks," *Applied spectroscopy*, vol. 74, no. 4, pp. 427–438, 2020.
- [24] M. T. Gebrekidan, C. Knipfer, and A. S. Braeuer, "Refinement of spectra using a deep neural network: Fully automated removal of noise and background," *Journal of Raman Spectroscopy*, vol. 52, no. 3, pp. 723–736, 2021.
- [25] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.
- [26] M. Kazemzadeh, C. L. Hisey, A. Artuyants, C. Blenkiron, L. W. Chamley, K. Zargar-Shoshtari, W. Xu, and N. G. Broderick, "Space curvature-inspired nanoplasmonic sensor for breast cancer extracellular vesicle fingerprinting and machine learning classification," *Biomedical Optics Express*, vol. 12, no. 7, pp. 3965–3981, 2021.
- [27] M. Kazemzadeh, C. L. Hisey, P. Dauros-Singorenko, S. Swift, K. Zargar-Shoshtari, W. Xu, and N. G. Broderick, "Label-free classification of bacterial extracellular vesicles by combining nanoplasmonic sensors with machine learning," *IEEE Sensors Journal*, 2021.
- [28] M. Kazemzadeh, M. Martinez-Calderon, S. Y. Paek, M. Lowe, C. Aguer-garay, W. Xu, L. W. Chamley, N. G. Broderick, and C. L. Hisey, "Classification of preeclamptic placental extracellular vesicles using femtosecond laser-fabricated nanoplasmonic sensors and machine learning," *bioRxiv*, 2021.
- [29] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [30] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *International conference on machine learning*. PMLR, 2015, pp. 448–456.

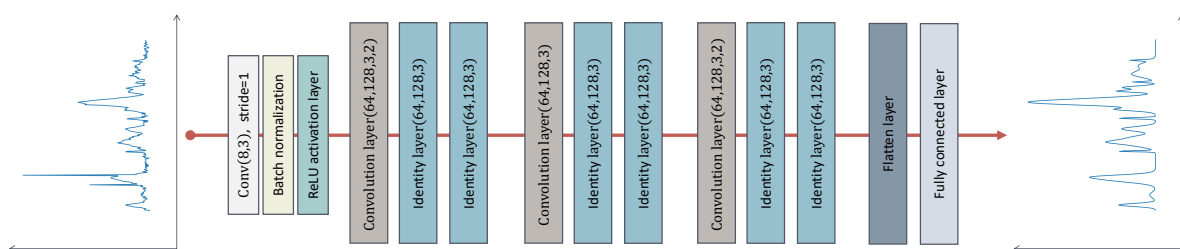
Supplementary Information: Deep Learning as an Improved Method of Preprocessing Biomedical Raman Spectroscopy Data

Mohammadrahim Kazemzadeh, Colin L. Hisey*, Miguel Martinez-Calderon, Lawrence W. Chamley, Weiliang Xu, *Senior-Member, IEEE*, Neil G.R. Broderick

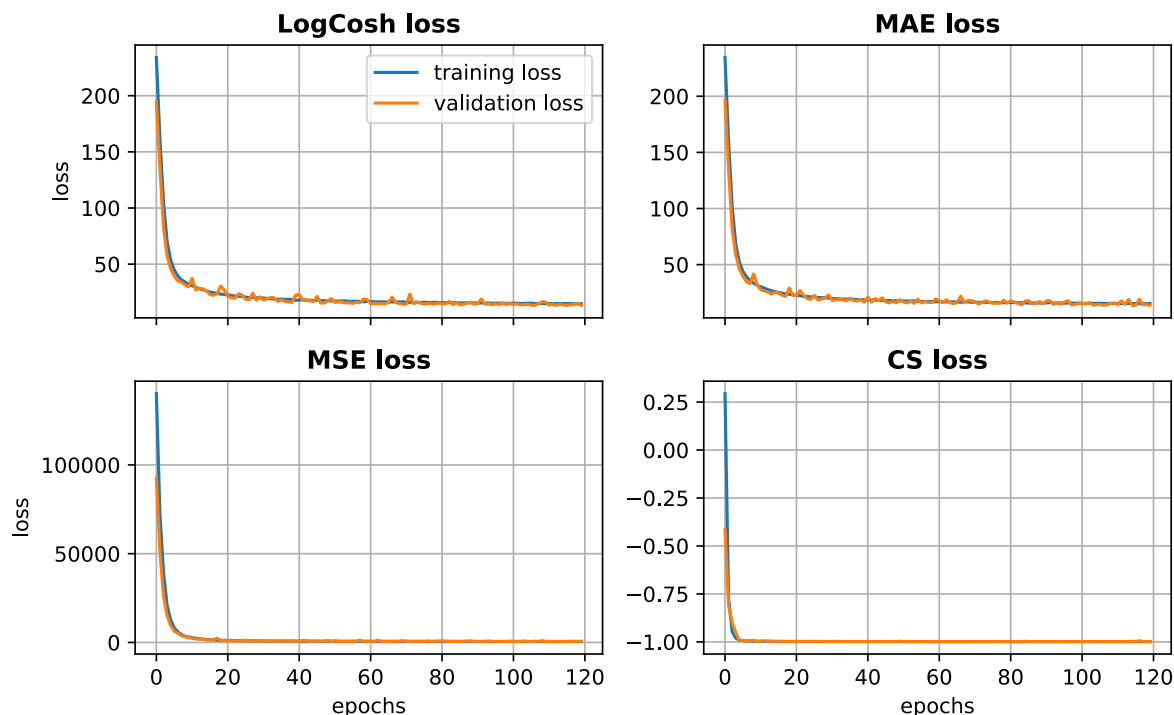
* Email: colin.hisey@auckland.ac.nz



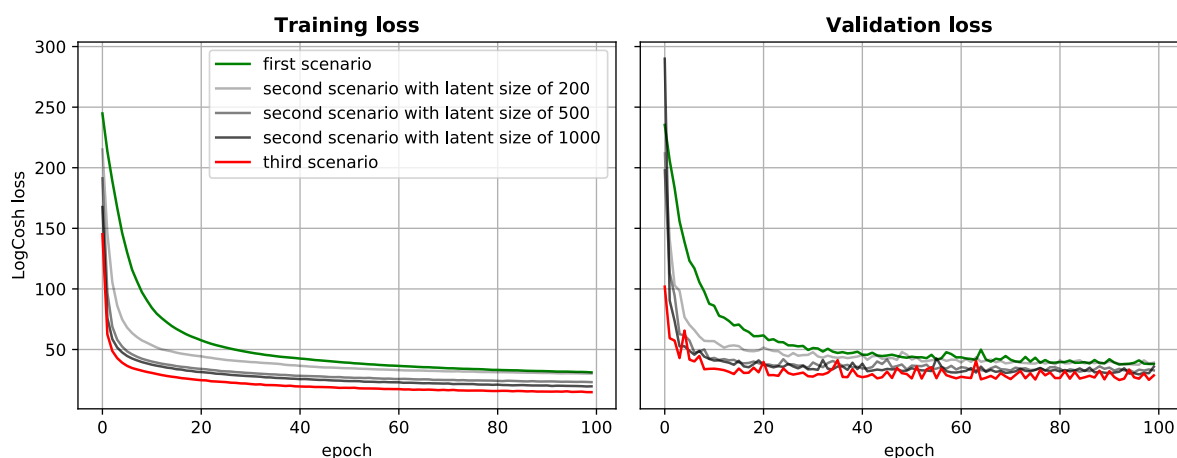
Supplementary Figure 1. Training and validation loss of the baseline removal network when Cosine Similarity (CS), Mean Absolute Error (MAE), Mean Square Error (MSE) and Logarithm of Cosine hyperbolic (LogCosh) losses are used.



Supplementary Figure 2. Network architecture for denoising.



Supplementary Figure 3. Training and validation loss of denoising network when Cosine Similarity (CS), Mean Absolute Error (MAE), Mean Square Error (MSE) and Logarithm of Cosine hyperbolic (LogCosh) losses are used.

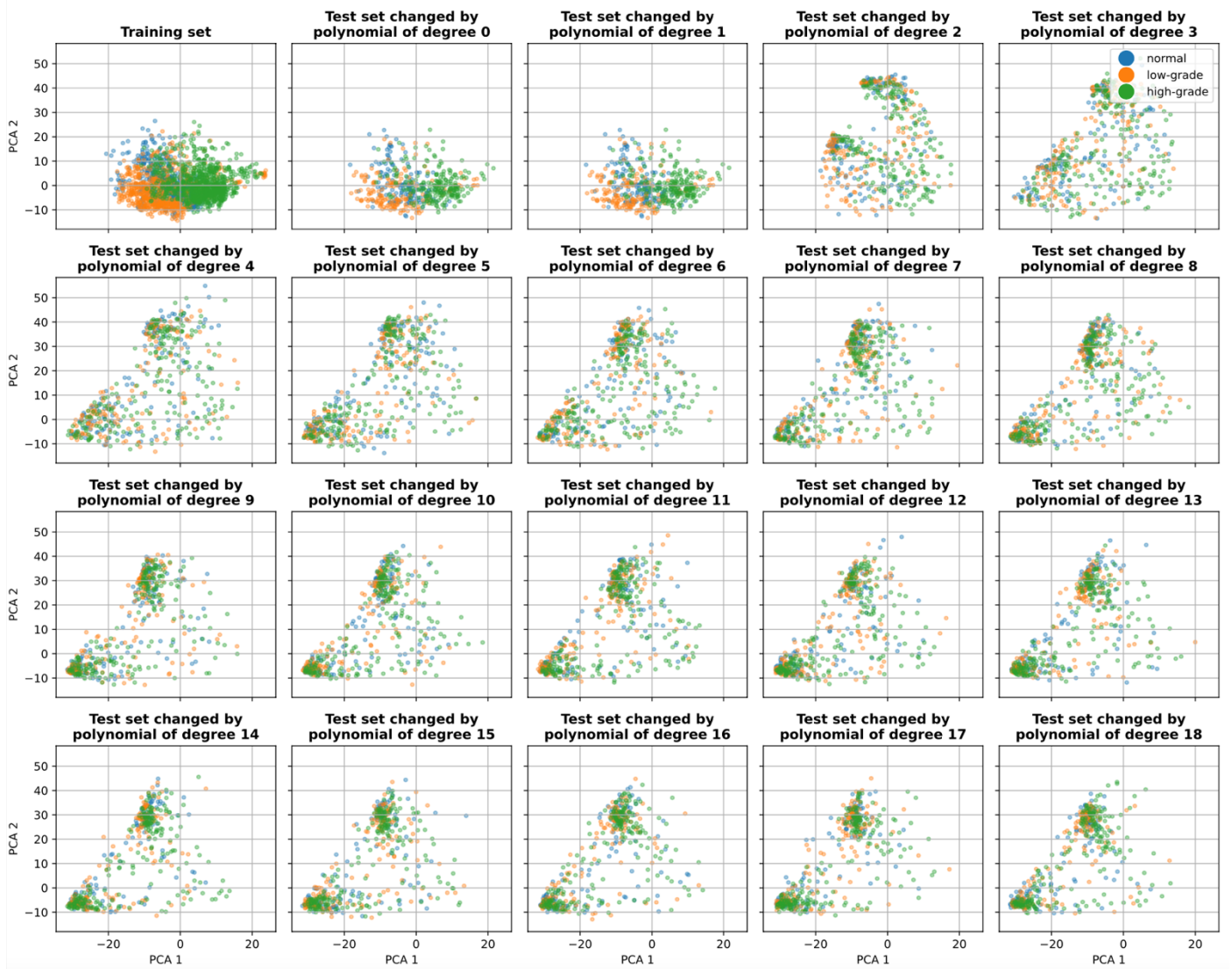


Supplementary Figure 4. Training and validation loss of networks doing all preprocessing steps including baseline removal and signal denoising for the first, second and third network architecture in Fig. 7.

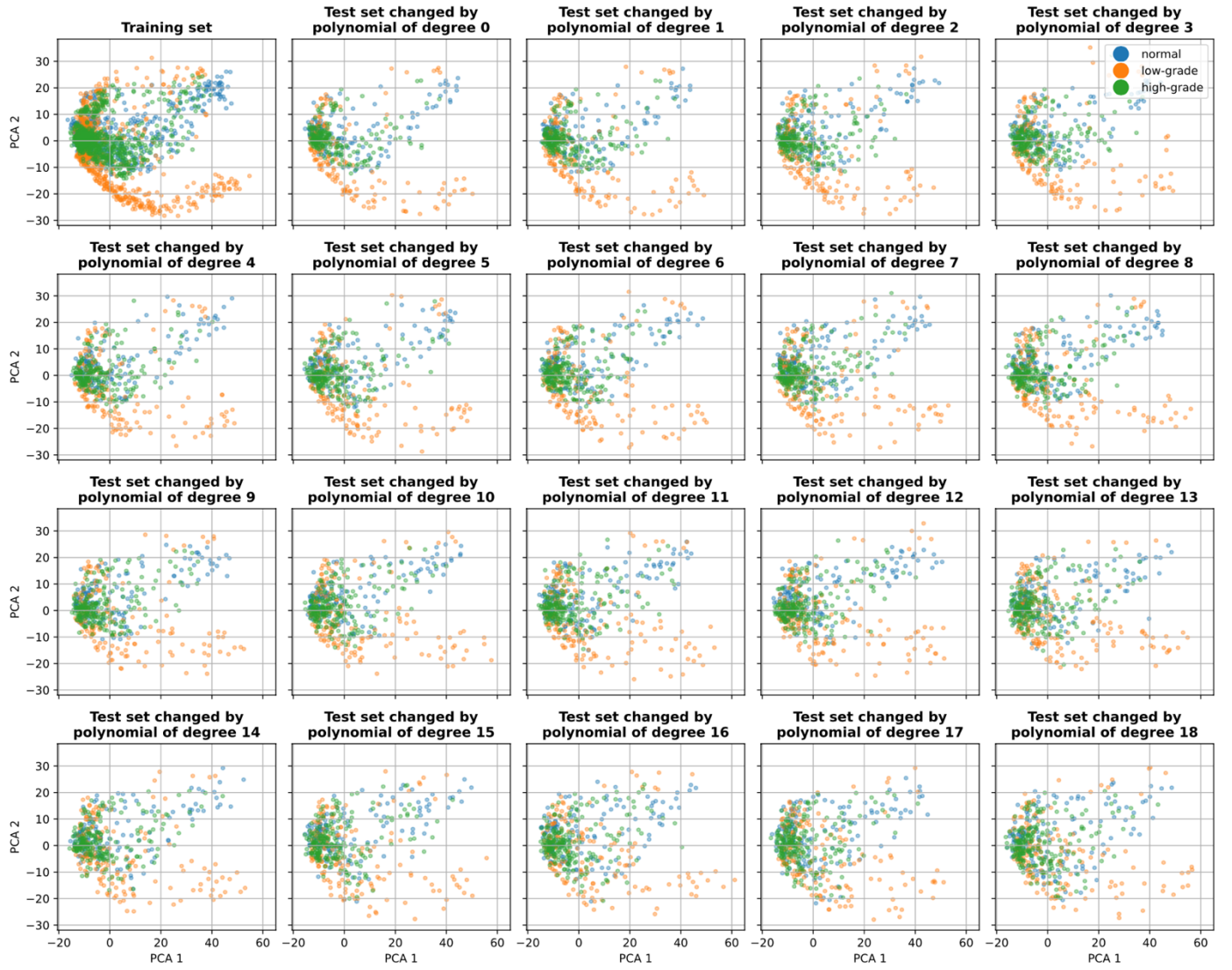
Additional Methods for Section IIIB:

Obtained classification accuracy between the data's labels was used as a measure to compare the performance of the different preprocessing techniques. To better see the effect of baselines and their removal, we also altered the test set's baselines using various polynomials. Then, by comparing the achieved accuracy when the training was done over the original data, we demonstrate how each preprocessing method can deal with different baselines. Moreover, this is also a simulation in which the data for the training and testing were obtained using different spectroscopy setup as different setups may lead to a different baselines and noise levels in their obtained spectra. The PCA transformation of the test set when the baselines of its spectra are altered with polynomials with degree between 0 to 18 and then corrected with AsLS combined with WL denoising and the proposed deep learning are shown in Supplementary Figure 5 and 6, respectively. Interestingly, this result (Fig S5) is in quite good agreement with the results of the synthesized data in Fig. 4, as the distribution of the spectra in the two-dimensional PCA plane start to deteriorate beyond

the second degree polynomial. In contrast, the proposed deep learning method shows much better performance even with much more complex baselines.



Supplementary Figure 5. Two dimensional PCA transformation of training set and the test set when the test set baselines are altered using various degrees of polynomials and then removed using asymmetric least square smoothing method (AsLS). The distribution of the spectra in a two dimensional PCA plain of the training set is almost replicated by the cases of added baseline with polynomial of zero degrees (without baseline) and one (straight line). For the polynomial of degree two and above, the distribution of test set deteriorates severely.



Supplementary Figure 6. Two dimensional PCA transformation of training set and the test set when the test set baselines are altered using various degree of polynomials and then removed using the proposed deep learning method. The distribution of the spectra in two dimensional PCA plains of the training set is replicated by all cases of added baselines to the test set.